NGUYEN KHAC BAO

Fullstack AI Integration Intern
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CAREER OBJECTIVE

AI integration intern candidate focused on turning AI models into usable products through backend APIs, RAG pipelines, vector databases, Docker, and cloud deployment. Hands-on with FastAPI, Streamlit, Gemini API, Qdrant Cloud, Cloud SQL, SSE-style streaming, and Cloud Run for deployable AI demos.

EDUCATION

Posts and Telecommunications Institute of Technology (PTIT) 2022 – 2027
Bachelor of Information Technology
Relevant Coursework: Machine Learning, Deep Learning, Computer Vision.

TECHNICAL SKILLS

AI Integration: RAG pipelines, Gemini API, embeddings, prompt engineering, OpenAI-compatible APIs, SSE-style streaming, fallback handling
Backend / API: Python, FastAPI, REST API, WebSocket, Pydantic
Databases: Qdrant Cloud, MySQL / Cloud SQL, SQL, JSON / JSONL, CSV processing
Cloud / DevOps: Docker, Google Cloud Run, Cloud Build, Artifact Registry, Secret Manager, Linux
ML / CV: PyTorch, YOLO, OpenCV, OCR, ByteTrack
Tools: Git / GitHub, Vast.ai (GPU), Hugging Face, Streamlit, Jupyter / Colab, Postman

LANGUAGES

Vietnamese: Native | English: TOEIC 550/990; technical reading proficiency for documentation, model cards, API references, and engineering articles.

PROJECTS

Vietnamese Legal RAG Chatbot – Fullstack AI App on Cloud Run 2026
Personal Project | Python, FastAPI, Streamlit, Gemini API, Qdrant Cloud, Cloud SQL/MySQL, Docker, Google Cloud Run | Source: GitHub | Demo: Cloud Run frontend/API health check

- Integrated Gemini as the default LLM and embedding provider, with Qdrant Cloud retrieval over a Vietnamese legal QA dataset for context-grounded answers.
- Built a cloud-ready fullstack RAG chatbot with a Streamlit frontend and FastAPI backend deployed as separate Google Cloud Run services.
- Used Cloud SQL/MySQL for chat history and Secret Manager/environment-based configuration for API keys, Qdrant credentials, and database credentials.
- Added SSE-style streaming response support, readiness checks, structured latency logging, and fallback handling for Qdrant/LLM failures without breaking the existing synchronous chat endpoint.

Real-Time Face Mask Compliance Detection – Deployed AI Inference Service 2026
Personal Project | Python, FastAPI, WebSocket, YOLOv8n, OpenCV, Docker, Google Cloud Run | Source: GitHub | Demo: Cloud Run webcam/API demo

- Built and deployed a real-time AI inference service with FastAPI REST + WebSocket endpoints and a browser webcam frontend running over HTTPS/WSS.
- Trained a YOLOv8n 3-class mask detector on PWMFD converted from VOC to YOLO format: 9,205 images, 18,528 labeled instances, and 0.970 mAP@50 validation.
- Logged warm endpoint latency around 151 ms for REST and 127 ms for WSS in the demo environment, with lightweight inference/FPS monitoring.
- Added violation event logging with per-event metadata (timestamp, confidence, snapshot) and a review API – mirroring camera-side alerting architecture.

Traffic Violation Detection & License Plate Recognition – AI Camera Evidence API 2026
Personal Project | Python, FastAPI, OpenCV, Ultralytics YOLO, ByteTrack, HyperLPR3, JSON | Source: GitHub

- Built a fixed-camera video analytics pipeline for vehicle detection/tracking, traffic-light ROI estimation, stop-line crossing logic, plate detection/OCR, and JSON evidence bundles.
- Built a FastAPI REST API exposing video analysis, event retrieval, and frame/plate evidence endpoints – enabling programmatic access to pipeline results for downstream integration.
- Prepared a compact CCPD2019 plate dataset with annotation parsing, YOLO bbox conversion, train/val/test split, negative samples, OCR crops, and QA reports.
- Fine-tuned a YOLO plate detector on 8,598 images, achieving 0.989 precision, 0.998 recall, and 0.994 mAP50 on validation; documented current OCR limitations and local demo constraints.

ACTIVITIES

AI Research Group – University Club 2025 – Present
Member
Discussed Machine Learning, Deep Learning, Computer Vision, RAG/LLM applications, and AI system development.